JACK NAIMER

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EDUCATION & TRAINING

Research Scholar Master of Computer Science – AI Specialization, EPFL (École Polytechnique Fédérale de Lausanne) 2025–2027
Awarded Research Scholarship for work in MLBio Lab led by Maria Brbic.

Bachelor of Applied Science in Engineering Science – Machine Intelligence Major, University of Toronto 2021–2025
Graduated with Honours. Received Certificate in Engineering Business.

Math and Computer Science (MaCS) enriched high school program, William Lyon Mackenzie Collegiate Institute 2017–2021
Achieved 5/5 on the college-level AP Computer Science exam in 10th grade.

PUBLICATIONS

"BioReason-Pro: Advancing Protein Function Prediction with Multimodal Biological Reasoning" – bioRxiv Preprint

"BioReason: Incentivizing Multimodal Biological Reasoning within a DNA-LLM Model" – Conference paper in The Annual Conference on Neural Information Processing Systems (NeurIPS).

"Semantically Safe Robot Manipulation: From Semantic Scene Understanding to Motion Safeguards" – Published in IEEE Robotics and Automation Letters.

RESEARCH & INDUSTRY EXPERIENCE

Master's Scholar Researcher, Dr. Maria Brbić, MLBio Lab, EPFL September 2025–Present
Developing a foundation model of single-cell chromatin accessibility (scATAC-seq).

Undergraduate Research Student, Dr. Bo Wang, University of Toronto/University Health Network March 2024–September 2025 (part-time)

- Ongoing development of BioReason-Protein, a protein-sequence model that combines protein-model embeddings with LLM-based reasoning to improve protein function prediction and interpretable prediction.
- Core BioReason team member; developed a DNA-LLM combining a DNA foundation model with an LLM to enable interpretable multi-step genomic reasoning.
- Led the development of an AI cardiac surgical scheduling system using tabular transformers and XGBoost; designed a 3-phase stepped wedge randomized controlled trial for evaluation and collaborated closely with senior hospital leadership.
- Honours Bachelor's thesis: Developed sparse-autoencoder methods to mechanistically interpret internal representations of the Evo genomic foundation model.

Undergraduate Researcher, Dr. Angela Schoellig Learning Systems and Robotics Lab (LSY Lab), Technical University of Munich, Germany May–August 2024
Developed AI-based robotic semantic safety filter. Used LLMs and VLMs to detect safety issues in a scene and generate robotic constraints.

Developed a construction and assembly robot using Franka robot arm. Focused on AI-vision components including part detection and pose tracking using foundation models.

Data Science Intern, Marsh, New York office June–August 2023
Developed a machine learning risk analysis and claims forecast model for hundreds of hospitals.

Research Student, SickKids Hospital with Dr. Sasha Litwin May–June 2023
Developed an AI chatbot prototype designed to answer a parent's fever questions, improve at-home treatment, and prevent unnecessary ER visits.

Research Student, Dr. Mike Fralick's Lab, Mount Sinai Hospital May 2022–May 2023
ChatBotMD: Created BERT-based AI chat bot prototype for clinical use, developed custom decision-making algorithms. Created chat bot website from scratch, set up networking, and run on AWS server with NGINX. Sole programmer.
Medical Literature Tool: Created website that uses my NLP and LLM approach to inform doctors of advances in medicine. Finds relevant papers and retrieves key information.
jrnowl.com: Using ML to automatically find journal publishing guidelines from medical journal websites for use on jrnowl.com.

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

Academic Director & Academic Paper Read Group Mentor, University of Toronto Machine Intelligence Student Team (UTMIST) 2022–2023

Software Team Member, University of Toronto Design League Robodog Team 2022
Responsible for inverse kinematics, and gait programming using ROS.

Competition Team Member, University of Toronto Concrete Toboggan Team 2021–2022

Satirical Newspaper Co-President, William Lyon Mackenzie CI 2020–2021

Math and Computer Science Program Peer Mentor & Calculus Tutor 2019–2021

TECHNICAL SKILLS

Python, Julia, C, C++, Java, Matlab, Linux, ROS, Docker, HTML, CSS, SolidWorks
Genomic foundation models, cell foundation models, protein foundation models
LLMs, NLP, Computer Vision, PyTorch, JAX, Tensorflow

AWARDS & ACHIEVEMENTS

Engineering Science Global Research Opportunities Program 2024
Highly competitive scholarship among Engineering Science students. For summer research in Angela Schoellig's LSY Lab in Munich.

BioTalent Health and Biosciences Student Work Placement Wage Subsidy 2022
For summer research in Dr. Mike Fralick's Lab.

Admission Scholarship, Faculty of Applied Science and Engineering, University of Toronto 2021
$2000

Advanced Placement (AP) 2019–2021
5/5 Computer Science A (2019), 5/5 Calculus AB (2021)